Radiology

Imaging of Osteoid Osteoma with Dynamic Gadolinium-enhanced MR Imaging[1]

Patrick T. Liu, MD
F. Spencer Chivers, MD
Catherine C. Roberts, MD
Christopher J. Schultz, MD
Christopher P. Beauchamp, MD

Index terms:
Computed tomography (CT), comparative studies
Magnetic resonance (MR), contrast enhancement
Osteoma, 45.3122

Published online
10.1148/radiol.2273020111
Radiology 2003; 227:691–700

[1] From the Departments of Radiology (P.T.L., F.S.C., C.C.R., C.J.S.) and Orthopedic Surgery (C.P.B.), Mayo Clinic Scottsdale, 13400 E Shea Blvd, Scottsdale, AZ 85259; and Radiology Medical Group of Napa, Calif (C.J.S.). From the 2001 RSNA scientific assembly. Received February 18, 2002; revision requested April 22; final revision received October 14; accepted November 6. **Address correspondence to** P.T.L. (e-mail: *liu.patrick@mayo.edu*).

Author contributions:
Guarantor of integrity of entire study, P.T.L.; study concepts, P.T.L., C.P.B.; study design, all authors; literature research, P.T.L.; clinical studies, P.T.L., C.C.R., F.S.C.; data acquisition, P.T.L., C.C.R., C.J.S., F.S.C.; data analysis/interpretation, P.T.L.; statistical analysis, all authors; manuscript preparation, definition of intellectual content, editing, revision/review, and final version approval, all authors.

© RSNA, 2003

PURPOSE: To compare dynamic gadolinium-enhanced T1-weighted magnetic resonance (MR) imaging with nonenhanced T1-weighted and T2-weighted MR imaging and thin-section computed tomography (CT) for the demonstration of osteoid osteomas.

MATERIALS AND METHODS: The images of 11 patients with pathologically proven osteoid osteomas who underwent nonenhanced MR imaging, dynamic gadolinium-enhanced MR imaging, and CT were retrospectively reviewed. Images obtained with all three techniques were scored for conspicuity of the osteoid osteoma relative to the surrounding bone. Time-enhancement curves were generated from signal intensity measurements of these lesions and the adjacent bone marrow. The mean imaging scores of the four techniques were compared, and the statistical significance was calculated by using a linear model with terms for method and patient. Pairwise comparisons were made by using the Tukey-Kramer adjustment for multiple comparisons.

RESULTS: Compared with CT, dynamic gadolinium-enhanced MR imaging demonstrated the osteoid osteoma equally well in eight of 11 patients and with better conspicuity in three of 11 patients, although this difference was not statistically significant ($P = .69$). The dynamic gadolinium-enhanced MR images demonstrated the osteoid osteomas significantly better than the nonenhanced T1-weighted ($P < .001$) and T2-weighted ($P < .001$) MR images. On the dynamic gadolinium-enhanced MR images, nine (82%) of 11 patients had peak enhancement of the osteoid osteoma in the arterial phase with early partial washout, compared with slower, progressive enhancement of the adjacent marrow. This resulted in greatest lesion to marrow contrast material enhancement in the arterial phase. One osteoid osteoma had peak enhancement in the venous phase, and one showed progressive enhancement through all phases to 150 seconds.

CONCLUSION: Osteoid osteomas can be imaged with greater conspicuity by using dynamic gadolinium-enhanced instead of nonenhanced MR imaging and with conspicuity equal to or better than that obtained with thin-section CT.
© RSNA, 2003

Osteoid osteomas usually manifest in people aged 10 through 20 years, with rare cases occurring in people younger than 5 years or older than 40 years (1). Most patients present with classical symptoms of bone pain that flares nocturnally but is promptly relieved with nonsteroidal anti-inflammatory drugs. The majority of the literature concerning osteoid osteomas describes computed tomography (CT) as being more accurate than magnetic resonance (MR) imaging at revealing the appearance of these lesions: an intracortical nidus, which is sometimes calcified, surrounded by medullary sclerosis and periosteal reaction (2–8). The finding of a double doughnut sign on radionuclide bone scans, both with and without single photon emission CT, has been found to be sensitive in the diagnosis of osteoid osteoma (9) but does not provide sufficient anatomic information to guide therapy.

Patients with symptoms of vague bone or joint pain, in whom the diagnosis of osteoid

TABLE 1
Patient Demographics, Lesion Description, and Time Interval between Studies

Patient No.	Age (y)	Lesion Location	Duration of Symptoms (mo)	Cortical or Intramedullary	Lesion Axial Diameter (mm)	Interval between Nonenhanced MR Imaging and Dynamic Gadolinium-enhanced MR Imaging (d)	Interval between Dynamic Gadolinium-enhanced MR Imaging and CT (d)	Use of Nonsteroidal Antiinflammatory Drugs at Time of MR
1	15	Left femoral middiaphysis	10	Cortical	5	0	0	No
2	26	T10 vertebra posterior element	4	Intramedullary	5	33	6	Yes
3	10	Right tibia proximal metaphysis	3	Cortical	5	0	0	No
4	29	Left supraacetabular ilium	48	Intramedullary	8	54	13	Yes
5	14	Left distal femoral metaphysis	5	Cortical	4	0	66	No
6	12	Right proximal femoral metaphysis	17	Cortical	8	0	15	Yes
7	17	Left femoral diaphysis	6	Cortical	5	0	0	No
8	11	Left distal diaphysis	5	Cortical	4	0	0	No
9	32	Left femoral neck	3	Cortical	5	0	21	Yes
10	14	Left distal tibial metaphysis	24	Intramedullary	4	0	0	No
11	11	Right femur proximal metaphysis	12	Cortical	5	0	0	No

Note.—All lesions had identical size measurements at CT and MR imaging.

osteoma is not initially suspected, often undergo MR imaging as their first cross-sectional imaging test. Patients with symptoms of classical osteoid osteoma might also undergo MR imaging if the results of CT are nonspecific. With the growing use of radio-frequency ablation in the treatment of osteoid osteoma during recent years, we have seen increasing referrals of both of these types of patients for CT and MR imaging. For these reasons, and to take advantage of the known hypervascularity of the tumor nidus, 2 years ago we began to use a dynamic, gadolinium-enhanced MR imaging sequence for imaging patients suspected of having osteoid osteomas.

The purpose of this retrospective study was to compare dynamic gadolinium-enhanced T1-weighted imaging with nonenhanced T1-weighted and T2-weighted MR imaging and thin-section CT scanning for the demonstration of osteoid osteomas.

MATERIALS AND METHODS

Patients

This retrospective review of patient images and records was approved by our institutional review board, and the requirement for patient informed consent was waived.

Two years ago, we began to use an MR imaging protocol in which dynamic gadolinium-enhanced MR imaging is used for examining patients suspected of having osteoid osteomas or small bone lesions that are difficult to characterize with routine nonenhanced T1-weighted and T2-weighted MR imaging sequences. All patients included in this review were examined with this protocol and were clinically referred for CT and MR imaging with histories of suspected osteoid osteoma, indeterminate small bone lesions on CT scans, or bone or joint pain.

One musculoskeletal radiologist (P.T.L.) searched our institution's MR imaging and CT records for patients who underwent imaging between January 2000 and December 2001 for suspected osteoid osteoma or had MR imaging findings that were suspicious for osteoid osteoma. Twenty-six patients met these criteria. Of these patients, we identified 11 with pathologically confirmed osteoid osteomas who underwent surgery or CT-guided radio-frequency ablation and nonenhanced MR imaging, dynamic gadolinium-enhanced MR imaging, and CT. These 11 patients constituted the study population. Mean patient age was 17.4 years (range, 11–32 years). Eight lesions were intracortical and three were intramedullary. Patient demographics and lesion locations are listed in Table 1. All patients were taking nonsteroidal anti-inflammatory drugs for pain control at the time of clinical presentation. Four patients continued taking nonsteriodal anti-inflammatory drugs through the time of MR imaging (Table 1).

The time intervals between nonenhanced MR imaging and dynamic gadolinium-enhanced MR imaging and between dynamic gadolinium-enhanced MR imaging and CT are also listed in Table 1. Dynamic gadolinium-enhanced MR imaging was usually performed on the same day as nonenhanced MR imaging, after review of the nonenhanced images. Two of the study patients underwent dynamic gadolinium-enhanced MR imaging several weeks after nonenhanced MR imaging because of scheduling constraints. In six patients, nonenhanced MR imaging and dynamic gadolinium-enhanced MR imaging were performed on the same day as CT as part of treatment planning evaluation. In others, CT was performed up to 9 weeks later.

Imaging Techniques

MR examinations were performed with either a 1.5-T Signa Echospeed imager (four patients) (GE Medical Systems, Milwaukee, Wis) or a 1.5-T ACS NT 6000 imager (seven patients) (Philips, Shelton, Conn) by using phased-array multicoils for imaging the pelvis, femur, and spine and by using extremity quadrature coils for imaging the knee and tibia. Imaging parameters are summarized in Table 2. The nonenhanced MR imaging techniques consisted of transverse T1-weighted spin-echo and fat-saturated T2-weighted fast spin-echo series, followed by repetition of the T2-weighted fat-saturated series in a longitudinal plane, with coronal or sagittal orientation chosen to best image the lesion in tangent.

TABLE 2
MR Imaging Parameters

Sequence	Flip Angle (°)	Repetition Time (msec)/ Echo Time (msec)	Bandwidth (kHz)	Echo Train*	Field of View (cm)	Section/Gap Thickness (mm)	Matrix	No. of Signals Acquired	Duration
T1-weighted spin echo†‡	90	500/10	20	NA	20–24	6/1	256 × 256	2	5 min 20 sec
T2-weighted fast spin echo, fat suppressed†‡	90	400/70	16	8	20–24	6/1	256 × 192	2	3 min 20 sec
T1-weighted fast field echo, dynamic fat suppressed‡	60	645/2.3	20	NA	20–24	6/1	256 × 160	2	1 min
T1-weighted fast spoiled gradient echo, suppressed, dynamic†	90	120/1.6	31.25	NA	20–24	6/1	256 × 160	2	1 min

* NA = not applicable.
† 1.5-T Signa Echospeed imager.
‡ 1.5-T ACS NT 6000 imager.

After review of the nonenhanced MR images, dynamic gadolinium-enhanced MR imaging was performed with a multisection two-dimensional T1-weighted fast spoiled gradient-echo sequence with the 1.5-T Signa Echospeed imager and a T1-weighted fast field-echo sequence with the 1.5-T ACS NT 6000 imager (see Table 2 for parameters). We already had extensive experience with this pulse sequence from our dynamic gadolinium-enhanced liver, kidney, and breast imaging protocols. Minor sequence modifications were made to increase image contrast and spatial resolution and resulted in longer imaging duration, which increased from 20 to 60 seconds. This image time was still thought to be acceptable for dynamic imaging in the extremities and spine, since breath holding was unnecessary while imaging these areas. These modifications included the addition of a fat-saturation prepulse to suppress the background marrow signal, a decrease of the field of view from 36 cm to 20–24 cm, a reduction of section thickness from 8 to 6 mm, addition of 2× phase oversampling, and an increase of signals acquired from one to two (to balance the signal loss incurred with the smaller voxel size). Sixteen to 20 sections were obtained during each contrast material–enhanced phase at 6-mm section thickness with 1-mm intersection gap. This relatively large volume was covered to examine for a possible second nidus and because, in some cases, the nidus could not be precisely localized with the preliminary nonenhanced MR imaging sequences.

An examination was performed prior to the administration of contrast material, followed by postcontrast examinations at 30, 90, and 150 seconds, which resulted in arterial, early venous, and late venous enhancement phases. As previously stated, the sequence duration was 60 seconds per phase. A power injector (Spectris; Medrad, Indianola, Pa) was used for intravenous administration of 0.1 mmol gadolinium chelate (Prohance; Bracco, Princeton, NJ) at 2 mL/sec; this was followed by a 20-mL normal saline flush. All imaging parameters remained identical for all dynamic gadolinium-enhanced MR imaging phases.

CT was performed with nonenhanced 2- or 3-mm-thick sections by using a Somatom Plus Four scanner (Siemens, Iselin, NJ). Parameters included 50% section overlap, 20–24-cm field of view, 200 mAs, 140 kVp, and reconstruction with bone algorithm.

Image Evaluation

Images of six adjacent sections from each technique (nonenhanced MR imaging, dynamic gadolinium-enhanced MR imaging, and CT) were chosen for filming, which ensured that the nidus was located in one of the central sections. A musculoskeletal fellowship–trained radiologist (P.T.L.) selected the images for review. The images from the three techniques were separated and interpreted in a randomized order. For statistical calculations, the nonenhanced MR images were separated into T1-weighted and T2-weighted images and scored individually. The four phases (one precontrast and three dynamic postcontrast) from dynamic gadolinium-enhanced MR imaging were all filmed at identical window and level settings, so any image brightness differences were due to contrast material enhancement. These images from the MR and CT examinations were reviewed retrospectively by two experienced musculoskeletal radiologists and one musculoskeletal radiology fellow (P.T.L., F.S.C., C.C.R.), who were blinded to the patient's name, surgical history, and pathologic information. Each reader reviewed and scored the images independently. When the readers' scores were in agreement, the scores were recorded as the final interpretation. When the reader's scores were discrepant, the images were reviewed again by all readers in a consensus panel, and these scores were recorded.

The imaging techniques were all subjectively graded for the conspicuity of the osteoid osteoma nidus on a scale of 1 to 4 (1 = none, 2 = poor, 3 = moderate, 4 = excellent). Readers were not given specific criteria for assigning each of the conspicuity grades. A training session was first performed with sample cases to demonstrate each level of conspicuity.

Enhancement Kinetics

To determine the enhancement kinetics of the osteoid osteomas, one reader (P.T.L.) measured the signal strength of the osteoid osteomas for all cases on a picture archiving and communication system workstation (Litebox; GE Medical Systems) with an ovoid region of interest that included as much of the enhancing nidus as possible while excluding the surrounding rim of sclerosis. Region-of-interest signal intensity was measured in the enhancing nidus three times at each dynamic phase, and the mean signal intensity value was used as the final signal intensity value.

Enhancement of bone marrow near the osteoid osteoma was also measured in the same manner on these same images at a distance of 1 cm from the osteoid osteoma with a round region of interest that measured 1 cm in diameter. Lesion-to–bone marrow signal intensity ratios were calculated at the four phases of enhancement.

Statistical Analysis

Interobserver reliability was calculated without consensus for each reader's scores.

The mean values of the final imaging scores for the four techniques were compared, and the statistical significance was calculated by using a linear model with terms for method and patient. Pairwise comparisons were made by using the Tukey-Kramer adjustment for multiple comparisons. A difference of at least 1 point was considered clinically significant.

RESULTS

Image Evaluation

The consensus imaging scores are summarized in Table 3. For all of the 11 osteoid osteomas, dynamic gadolinium-enhanced MR imaging provided increased conspicuity of the tumor nidus compared with nonenhanced MR imaging ($P < .001$). In three of 11 patients, the osteoid osteoma was more conspicuous on the dynamic gadolinium-enhanced MR images than on the CT images. One of these cases is shown in Figure 1. In the eight remaining patients, the nidus was equally well demonstrated with both dynamic gadolinium-enhanced MR imaging and CT (Figs 2, 3). In no case was the CT score higher than the dynamic gadolinium-enhanced MR imaging score.

When we examined the individual reviewers' scores (without consensus) for the four imaging techniques in these 11 patients, the reliability between reviewers was 0.70 for T1-weighted MR imaging (95% CI: 0.39, 0.90), 0.54 for T2-weighted MR imaging (95% CI: 0.17, 0.83), 0.61 for CT (95% CI: 0.25, 0.86), and 0.02 for dynamic gadolinium-enhanced MR imaging (95% CI: −0.26, 0.47). While these κ values show high interobserver variability for scoring of the dynamic gadolinium-enhanced MR images, 32 (97%) of 33 of these individual scores rated dynamic gadolinium-enhanced MR imaging equal to or better than CT, and 32 (97%) of 33 scores rated lesion conspicuity at dynamic gadolinium-enhanced MR imaging as either 3 (moderate) or 4 (excellent).

TABLE 3
Scores for Conspicuity of Osteoid Osteoma on Images Obtained with Thin-Section CT, Nonenhanced MR, and Dynamic Gadolinium-enhanced MR

Patient No.	Imaging Technique			
	Thin-Section CT	Nonenhanced MR Imaging		Dynamic Gadolinium-enhanced MR Imaging
		T1-weighted	T2-weighted	
1	4	3	3	4
2	2	2	2	3
3	3	2	3	4
4	4	2	3	4
5	4	2	2	4
6	3	2	2	4
7	4	3	3	4
8	4	3	3	4
9	3	2	2	3
10	4	3	4	4
11	4	3	2	4
Mean*	3.6 (0.69)	2.5 (0.52)	2.6 (0.67)	3.8 (0.40)

* Numbers in parentheses are SDs.

Enhancement Kinetics

Time-enhancement curves were created by plotting percentage enhancement of the osteoid osteomas relative to baseline against the delay time from the injection of the contrast material to the start of the imaging phase (Fig 4). These time-enhancement curves demonstrate that all lesions enhanced rapidly in the arterial phase. Nine (82%) of the 11 tumors had peak enhancement in the arterial phase (30 seconds), with rapid washout of contrast material between the arterial and early venous (90 seconds) phases. One nidus had a peak at the early venous phase, with washout at the late venous phase. Patient 9 showed continually increased enhancement until 150 seconds.

Figure 5 shows the time-enhancement curves for bone marrow adjacent to the osteoid osteomas of the 11 subjects. The marrow showed progressively increasing enhancement during the measurement period for nine (82%) of 11 patients with a slower rate than the osteoid osteomas. Patient 2 had an early peak of marrow enhancement at 30 seconds followed by slight washout, while patient 9 showed only minimal fluctuation of marrow signal intensity.

Figure 6 is the graph of lesion-to-marrow enhancement ratio during the four dynamic gadolinium-enhanced MR imaging phases. The plot shows that the lesion-to-marrow enhancement ratio increased rapidly, peaking at the 30 second mark for nine (82%) of the 11 patients, before returning nearly to the baseline ratio at 150 seconds. For patient 9, the curve shows continuous growth of the lesion-to-marrow enhancement ratio during the entire measurement period. For patient 2, the greatest lesion-to-marrow enhancement ratio occurred at 90 seconds before it dropped at 150 seconds, but this was apparently due to the small washout of marrow contrast material at 90 seconds, and it is not reflected on the time-enhancement curve (Fig 5).

Examples of the nonenhanced MR, dynamic gadolinium-enhanced MR, and CT images are shown for three of the patients in this study. Patients 6 (Fig 1), 4 (Fig 2), and 5 (Fig 3) all presented with symptoms that were atypical for osteoid osteoma and were referred for MR imaging after radiographs were obtained and found to be nonrevealing. Dynamic gadolinium-enhanced MR imaging was performed in patients 6 and 4 after the radiologists reviewed the nonenhanced MR images and observed the suspicious lesions. An orthopedist referred patient 5 for dynamic gadolinium-enhanced MR imaging after identifying focal subcortical marrow edema on an earlier nonenhanced MR image obtained elsewhere (images not available).

The readers' scores indicated that the dynamic gadolinium- enhanced MR image demonstrated the osteoid osteomas with greater conspicuity than the nonenhanced MR images for these three patients. The readers believed that dynamic gadolinium-enhanced MR imaging was equal to CT in patients 4 and 5 but had greater conspicuity than CT in patient 6.

According to the Tukey-Kramer analysis, the slightly higher dynamic gadolinium-enhanced MR imaging scores did not significantly differ from the CT scores (Table 4); however, dynamic gadolinium-enhanced MR imaging scores were significantly higher than those assigned to the T1-weighted and T2-weighted techniques. CT was also scored significantly higher than the T1-weighted MR technique. There was no significant difference be-

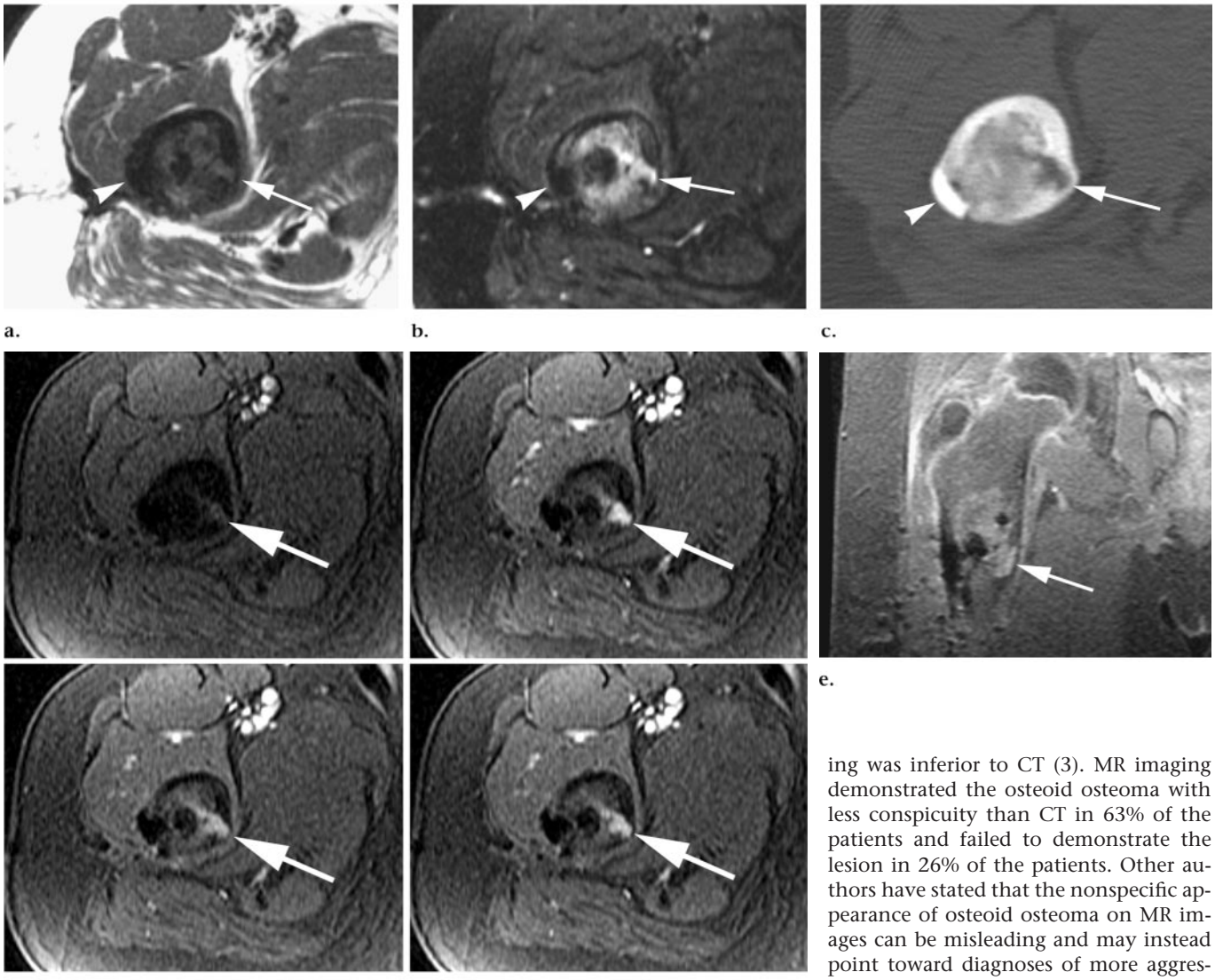


Figure 1. Patient 6. A 12-year-old boy with intermittent right hip pain who previously underwent biopsy and curettage of proximal femoral metaphysis marrow lesion found at MR imaging performed elsewhere (not shown). His pain persisted despite the surgery. Transverse **(a)** T1-weighted and **(b)** T2-weighted fat-suppressed MR images show a lateral bone graft (arrowhead), diffusely decreased T1-weighted signal intensity and increased T2-weighted signal intensity in the medullary space, and a medial 7-mm cortical defect (arrow). **(c)** Transverse CT image shows that the medial cortical lesion is hypoattenuating compared with surrounding bone (arrow), which suggests a differential diagnosis of surgical defect, Brodie abscess, or osteoid osteoma. Cortical bone graft (arrowhead) is visible lateral to the femur. **(d)** Transverse dynamic gadolinium-enhanced MR images obtained (clockwise from upper left) before injection of contrast material and 30, 90, and 150 seconds after injection of contrast material show rapid enhancement of the focal lesion (arrow) in the medial cortex, followed quickly by partial washout. **(e)** Delayed coronal T1-weighted fast spoiled gradient-echo image obtained 5 minutes after injection of contrast material with the same parameters as dynamic gadolinium-enhanced MR imaging. Image shows that the osteoid osteoma (arrow) and adjacent marrow have similar signal intensity. Biopsy revealed osteoid osteoma. The patient's pain was relieved with CT-guided radio-frequency ablation. Dynamic gadolinium-enhanced MR imaging was scored as better than nonenhanced MR imaging and CT.

tween the scores for the CT and the scores for the T2-weighted MR techniques and between the scores for the T1-weighted MR and the scores for the T2-weighted MR techniques.

DISCUSSION

In a 1994 study, Assoun et al compared MR imaging with CT of osteoid osteoma in 19 patients and found that MR imaging was inferior to CT (3). MR imaging demonstrated the osteoid osteoma with less conspicuity than CT in 63% of the patients and failed to demonstrate the lesion in 26% of the patients. Other authors have stated that the nonspecific appearance of osteoid osteoma on MR images can be misleading and may instead point toward diagnoses of more aggressive tumors or infection (4,5,7). In a more recent study, however, Spouge and Thain (10) found that MR imaging performed with higher spatial resolution techniques demonstrated a nidus in 10 of 10 patients with osteoid osteoma and outperformed CT in two patients whose tumors were located in cancellous bone.

These and earlier MR imaging studies and case series of osteoid osteomas focused mainly on findings that were visible on images obtained with nonenhanced sequences, such as high signal intensity on T2-weighted images of the nidus and periosteal and bone marrow edema (3–7,10–13). The finding that the nonenhanced MR imaging techniques used in these earlier studies could not reliably depict the osteoid osteomas was probably due to a combination of insufficient spatial resolution and obscuration by surrounding marrow and periosteal edema. When gadolinium-enhanced MR imaging was used in these earlier studies,

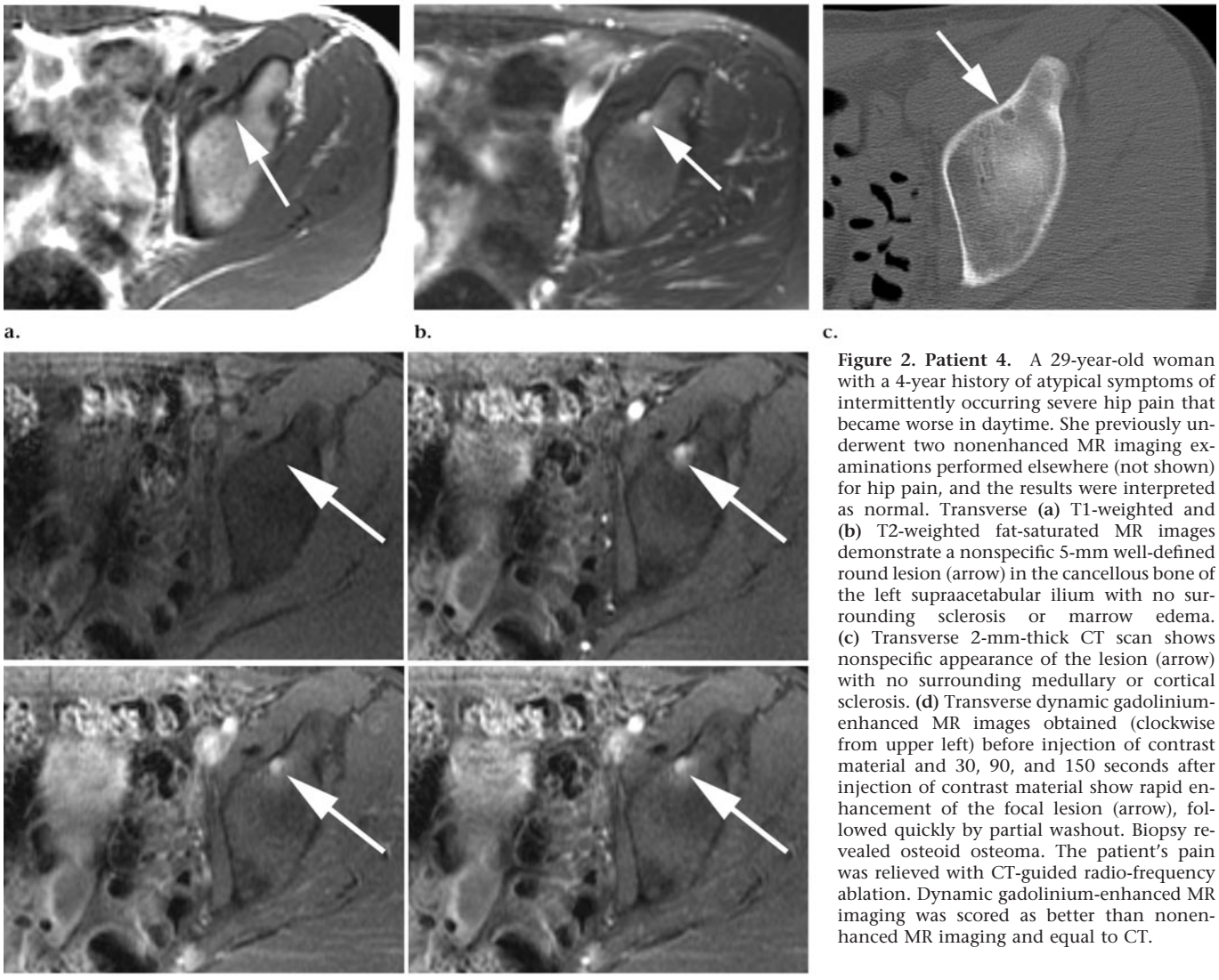


Figure 2. Patient 4. A 29-year-old woman with a 4-year history of atypical symptoms of intermittently occurring severe hip pain that became worse in daytime. She previously underwent two nonenhanced MR imaging examinations performed elsewhere (not shown) for hip pain, and the results were interpreted as normal. Transverse **(a)** T1-weighted and **(b)** T2-weighted fat-saturated MR images demonstrate a nonspecific 5-mm well-defined round lesion (arrow) in the cancellous bone of the left supraacetabular ilium with no surrounding sclerosis or marrow edema. **(c)** Transverse 2-mm-thick CT scan shows nonspecific appearance of the lesion (arrow) with no surrounding medullary or cortical sclerosis. **(d)** Transverse dynamic gadolinium-enhanced MR images obtained (clockwise from upper left) before injection of contrast material and 30, 90, and 150 seconds after injection of contrast material show rapid enhancement of the focal lesion (arrow), followed quickly by partial washout. Biopsy revealed osteoid osteoma. The patient's pain was relieved with CT-guided radio-frequency ablation. Dynamic gadolinium-enhanced MR imaging was scored as better than nonenhanced MR imaging and equal to CT.

the enhanced MR imaging was performed in a static nondynamic mode. Timing information was not given in these studies, but the postcontrast images were most likely acquired at least several minutes after administration of contrast material. These enhanced images were not thought to give much added information.

In our study, we demonstrated that dynamic gadolinium- enhanced MR imaging can depict osteoid osteomas with greater conspicuity than nonenhanced MR imaging techniques ($P < .001$) and with equal conspicuity compared with thin-section CT. To the best of our knowledge, no previous study has examined the use of dynamic gadolinium-enhanced MR imaging for osteoid osteomas, a technique that is routinely used for contrast-enhanced multiphase imaging in the liver, pancreas, kidneys, and breast. It is well recognized that some hypervascular lesions in these organs will be visible only on the arterial phase images, before they wash out and become isointense with the surrounding background tissue. Hypervascular tumors often lose conspicuity during the later phases of enhancement at dynamic gadolinium-enhanced MR imaging due to two factors: the rapid washout of the contrast material because of fast blood flow and the slower enhancement of the surrounding background tissue. These same principles are likely responsible for the ability of dynamic gadolinium-enhanced MR imaging to demonstrate the osteoid osteoma better than nonenhanced MR imaging.

The time-enhancement curves we obtained show that the majority of osteoid osteomas had arterial phase enhancement and rapid partial washout (Fig 4). Only two of the 11 cases lacked this characteristic enhancement pattern: One case showed peak enhancement in the early venous phase, and the other showed progressively increasing enhancement throughout all phases, although both cases showed some enhancement in the arterial phase. We speculate that the vascularity of these lesions differed somewhat from the remainder of our study patients; however, the pathology reports from the corresponding biopsy specimens did not address this issue.

In the patients with rapid washout of the osteoid osteomas, we believe that postcontrast imaging with a slower spin-echo sequence would have been less sensitive than dynamic imaging with a fast

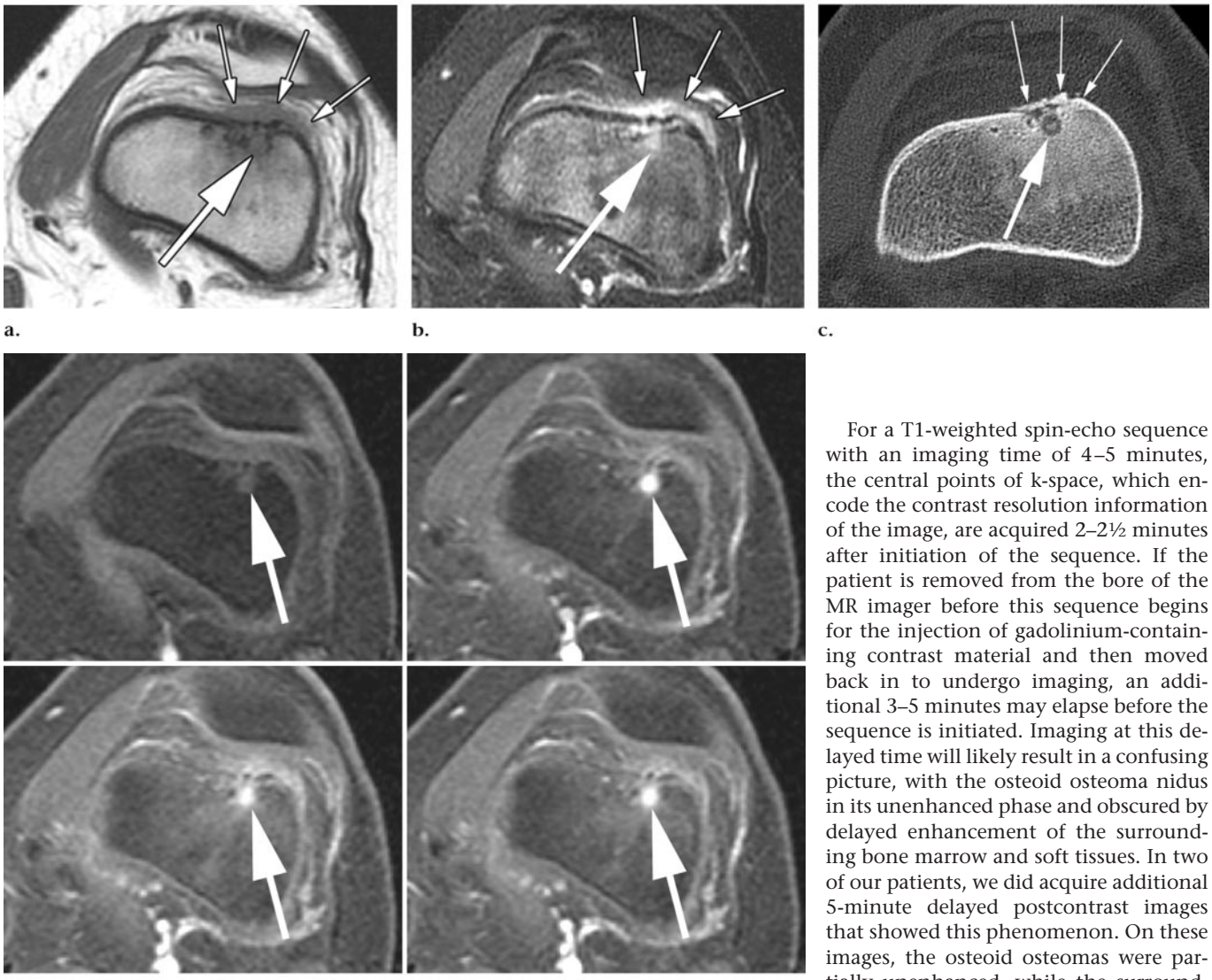


Figure 3. Patient 5. A 14-year-old girl with a 5-month history of intermittent knee pain throughout the day and night. Initial radiographs and nonenhanced MR images obtained elsewhere (not shown) were interpreted as normal. Transverse **(a)** T1-weighted and **(b)** T2-weighted fat-suppressed MR images of the distal femoral metaphysis show a 1-cm anterior subcortical lobulated lesion (thick arrow) surrounded by a poorly defined area of cortical and subcortical edema and periosteal reaction (thin arrows). **(c)** Transverse CT image shows a 4-mm-diameter, round, hypoattenuating, subcortical lesion (thick arrow) with a calcified nidus and surrounding medullary sclerosis. Overlying cortical disruption and periosteal reaction (thin arrows) indicate possible Brodie abscess or osteoid osteoma. **(d)** Transverse dynamic gadolinium-enhanced MR images obtained (clockwise from upper left) before injection of contrast material and 30, 90, and 150 seconds after injection of contrast material show rapid enhancement of the focal lesion (arrow), followed quickly by partial washout. Early enhancement is localized to the nidus, and there is delayed enhancement of the surrounding marrow and periosteal edema. Biopsy of the enhancing lesion was performed and revealed osteoid osteoma. The patient's pain was relieved with CT-guided radio-frequency ablation. Dynamic gadolinium-enhanced MR imaging was scored as better than nonenhanced MR imaging and equal to CT.

gradient-echo sequence such as the one used in this study. The difference between the enhancement rates of osteoid osteomas and the adjacent bone marrow can be seen by comparing Figures 4 and 5. Figure 6 indicates that the arterial phase enhanced image has the greatest likelihood of demonstrating the enhancing osteoid osteoma nidus. The combination of the osteoid osteoma washout and the delayed enhancement of the marrow will likely result in decreased conspicuity of the lesion on the images obtained with delayed sequences.

For a T1-weighted spin-echo sequence with an imaging time of 4–5 minutes, the central points of k-space, which encode the contrast resolution information of the image, are acquired 2–2½ minutes after initiation of the sequence. If the patient is removed from the bore of the MR imager before this sequence begins for the injection of gadolinium-containing contrast material and then moved back in to undergo imaging, an additional 3–5 minutes may elapse before the sequence is initiated. Imaging at this delayed time will likely result in a confusing picture, with the osteoid osteoma nidus in its unenhanced phase and obscured by delayed enhancement of the surrounding bone marrow and soft tissues. In two of our patients, we did acquire additional 5-minute delayed postcontrast images that showed this phenomenon. On these images, the osteoid osteomas were partially unenhanced, while the surrounding edematous marrow had mild enhancement, similar in degree to the osteoid osteomas (Fig 1e). These enhancement kinetics may explain the lack of sensitivity reported in prior studies that used delayed contrast-enhanced MR imaging of osteoid osteomas.

In every case, we believed dynamic gadolinium-enhanced MR imaging techniques demonstrated the osteoid osteomas with greater conspicuity than either of the nonenhanced MR imaging techniques. Possible explanations for this difference include the presence of associated marrow edema, periosteal edema or new bone formation, marrow sclerosis, cortical thickening, or calcification of the nidus obscuring the lesion to varying degrees on the nonenhanced MR images. These findings can obscure the margins of the lesion and make it appear larger and more aggressive, and they have all been cited as possible causes for the non-

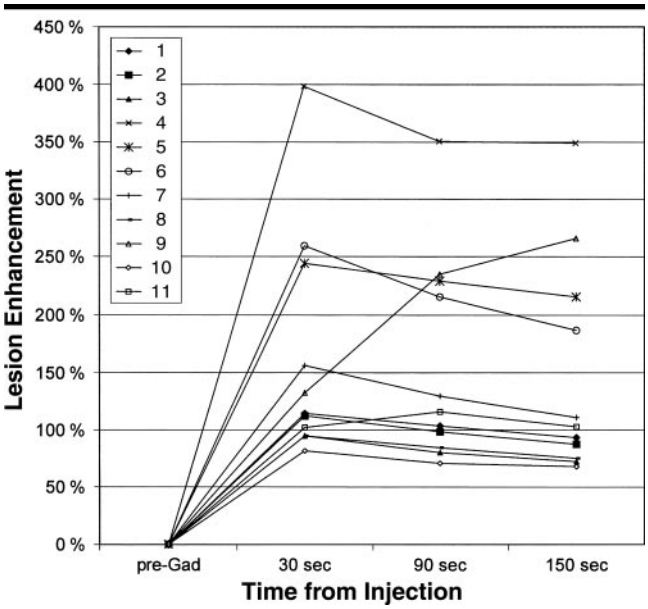


Figure 4. Time-enhancement graph for lesions. Nine of the 11 patients showed peak enhancement at 30 seconds followed by rapid partial washout. Patient 11 showed atypical enhancement peak at 90 seconds, and patient 9 showed progressive enhancement throughout all phases. Numbers in box refer to patient numbers. *pre-Gad* = before administration of contrast material.

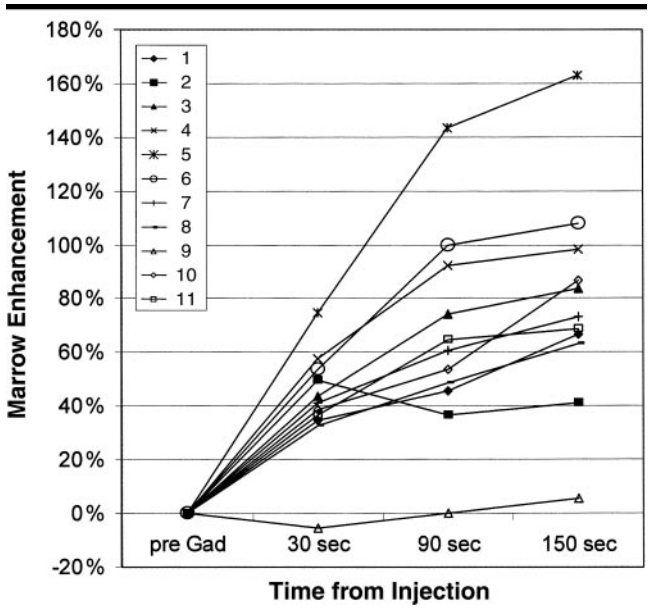


Figure 5. Time-enhancement graph for bone marrow adjacent to the osteoid osteomas. Signal intensity of bone marrow was measured at a distance of 1 cm from the osteoid osteoma nidus. Marrow enhancement generally increased progressively throughout all imaging phases. Numbers in box refer to patient numbers. *pre-Gad* = before administration of contrast material.

specific performance of MR imaging for detection of osteoid osteomas (2–5,7,14).

Dynamic gadolinium-enhanced MR imaging, however, makes use of the physiologic characteristics of the osteoid osteoma nidus. Hypervascularity of the nidus on angiograms has been reported in several previous case series (15–17), and histologic examination of osteoid osteomas has revealed hypervascular stroma (18). Two preliminary studies of dynamic contrast-enhanced CT took advantage of this hypervascularity to show that it was useful for differentiating between an osteoid osteoma and other similar small radiolucent bone lesions (19,20). Both of these studies showed prominent peak enhancement during the arterial phase and rapid washout in the histologically proven osteoid osteomas in comparison with mild delayed phase enhancement in their cases of Brodie abscess, eosinophilic granuloma, and chondroblastoma.

Dynamic CT, however, exposes the patient to a greater radiation dose than routine single-phase CT. Nearly all patients with osteoid osteomas are less than 40 years old, and many are less than 20 years old. In this age range, the potential harm of ionizing radiation is worrisome and can be avoided entirely with MR. The enhancement of the tumor nidus relative to the background tissue can also be made more obvious with MR imaging than with CT, because fat suppression can be used with MR to eliminate nearly all signal intensity from the bone marrow. On CT images, the nidus sometimes appears partially calcified and is commonly surrounded by dense, sclerotic bone cortex, potentially making contrast enhancement less obvious.

Dynamic MR imaging of other bone tumors and soft-tissue sarcomas has proven useful in the evaluation of tumor viability after chemotherapy (21,22). These previous studies used single-section multiphase imaging, which enabled the use of very fast examinations at a higher temporal resolution than in our study. We chose to use a pulse sequence that permitted more section coverage, which enabled us to evaluate the surrounding bone and soft tissues. This larger imaging volume may be needed in cases in which the location of an osteoid osteoma nidus cannot definitively be determined with CT scanning or nonenhanced MR imaging.

The use of a gradient-echo sequence in the evaluation of bone lesions has been discouraged due to magnetic susceptibility artifact. This artifact is generated by the bone trabeculae and cortex and can obscure bone lesions on longer echo time images (10). We attempted to minimize this artifact with the fast spoiled gradient-echo sequence we used for the dynamic gadolinium-enhanced MR imaging by using very short (1.6–2.3-msec) echo times.

Targeted CT performed with thin sections and a small field of view remains an excellent test to aid in the diagnosis and localization of osteoid osteoma when the clinical history is classical. We do not suggest that dynamic gadolinium-enhanced MR imaging should replace CT as the preferred choice of cross-sectional imaging for patients suspected of having osteoid osteomas; rather, we suggest that those lesions that are believed to be indeterminate or suspicious for osteoid osteoma at nonenhanced MR imaging or CT can be more specifically imaged with the addition of a dynamic gadolinium-enhanced MR imaging sequence. The demonstration of an unsuspected nidus in an area of bone sclerosis or edema may lead the radiologist to the diagnosis of osteoid osteoma.

Limitations of our study include retrospective design and small sample size. We did not examine the appearance of dynamic gadolinium-enhanced MR images in patients with other types of bone lesions, so we cannot assess the sensitivity and specificity of this technique; however, in simple bone cysts and Brodie abscesses, lesions that may have a radiologic appearance similar to that of an osteoid osteoma, we expect that the contrast-enhanced images would show central hypo-

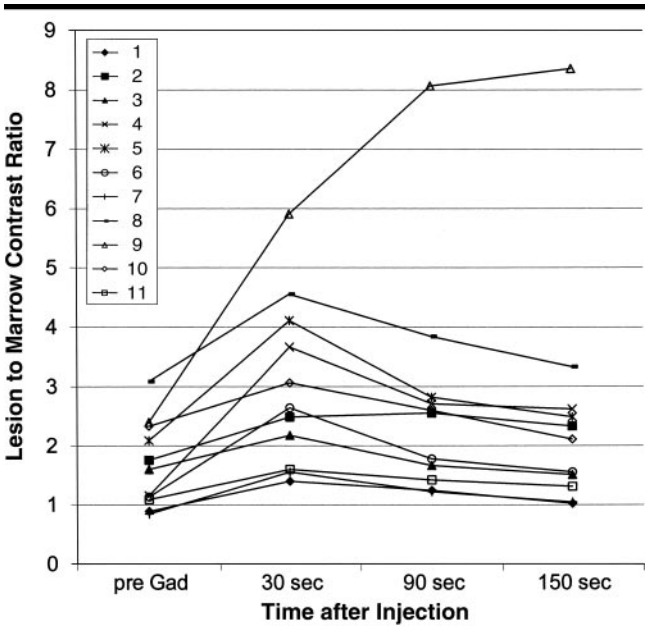


Figure 6. Graphs of lesion-to-marrow enhancement ratio. For most patients, the peak lesion-to-marrow enhancement occurred at 30 seconds and returned nearly to the baseline level by 150 seconds. Numbers in box refer to patient numbers. *pre-Gad* = before administration of contrast material.

TABLE 4
Tukey-Kramer Comparison of Imaging Scores

Techniques Compared	Mean ± SD Tukey-Kramer Score	*P* Value	95% CI
Dynamic gadolinium-enhanced MR imaging vs CT	0.27 ± 0.47	.69	−0.39, 0.93
Dynamic gadolinium-enhanced MR imaging vs T2-weighted MR imaging	1.18 ± 0.60	<.001	0.52, 1.84
Dynamic gadolinium-enhanced MR imaging vs T1-weighted MR imaging	1.36 ± 0.50	<.001	0.70, 2.02
CT vs T2-weighted MR imaging	0.91 ± 0.70	.003	0.25, 1.57
CT vs T1-weighted MR imaging	1.09 ± 0.54	<.001	0.43, 1.75
T2-weighted MR imaging vs T1-weighted MR imaging	0.18 ± 0.60	.88	−0.48, 0.84

Note.—A mean score of greater than 1 indicates a significant difference between the compared techniques.

vascularity within the lesions. This expectation is based on our prior experience with MR imaging of intraosseous abscesses and bone cysts and on the reported lack of central enhancement in a Brodie abscess at dynamic CT (20).

While the κ scores for the CT and the T1-weighted and T2-weighted MR techniques were good, the reliability for dynamic gadolinium-enhanced MR was poor. This may be due either to the subjective scoring method used in this study or to observer bias. It is helpful to note however, that 32 (97%) of the 33 individual scores for the dynamic gadolinium-enhanced MR images were higher than those for the corresponding CT images.

The variability and length of time between the three types of imaging examinations is another source of potential error, but this could not be controlled due to the retrospective nature of the study. Osteoid osteomas are very slowly growing benign tumors that are unlikely to change substantially, if at all, within a space of several months. The tumors in this series showed no growth during the time of the study. Most patients with these tumors will have had symptoms for several months when their tumors are diagnosed, and nearly all osteoid osteomas, including those in our series, are less than 1 cm in diameter at the time of diagnosis.

It should also be recognized that, with this study design, observer bias could not be eliminated since the reviewers could not be blinded to the type of image they were evaluating. Additional prospective studies are needed to determine the true sensitivity and specificity of the dynamic gadolinium-enhanced MR imaging technique for the diagnosis of osteoid osteoma.

In conclusion, we have shown that dynamic gadolinium-enhanced MR imaging can depict osteoid osteomas with greater conspicuity than nonenhanced MR imaging and with conspicuity equal to that of thin-section CT. Thus, MR imaging performed with a dynamic gadolinium-enhanced sequence can provide valuable added information in the imaging workup of osteoid osteoma. Dynamic gadolinium-enhanced MR imaging may also be helpful in cases in which CT images show findings that suggest the presence of an osteoid osteoma.

Acknowledgments: Thanks to Joseph G. Hentz, MS, for assistance with statistical analysis and to Ethan Braunstein, MD, for editorial assistance.

References

1. Kransdorf MJ, Stull MA, Gilkey FW, Moser RP Jr. Osteoid osteoma. RadioGraphics 1991; 11:671–696.
2. Hayes CW, Conway WF, Sundaram M. Misleading aggressive MR imaging appearance of some benign musculoskeletal lesions. RadioGraphics 1992; 12:1119–1134; discussion 1135–1136.
3. Assoun J, Richardi G, Railhac JJ, et al. Osteoid osteoma: MR imaging versus CT. Radiology 1994; 191:217–223.
4. Goldman AB, Schneider R, Pavlov H. Osteoid osteomas of the femoral neck: report of four cases evaluated with isotopic bone scanning, CT, and MR imaging. Radiology 1993; 186:227–232.
5. Woods ER, Martel W, Mandell SH, Crabbe JP. Reactive soft-tissue mass associated with osteoid osteoma: correlation of MR imaging features with pathologic findings. Radiology 1993; 186:221–225.
6. Okuda S, Myoui A, Nakase T, Wada E, Yonenobu K, Yoshikawa H. Ossification of the ligamentum flavum associated with osteoblastoma: a report of three cases. Skeletal Radiol 2001; 30:402–406.
7. Lefton DR, Torrisi JM, Haller JO. Vertebral osteoid osteoma masquerading as a malignant bone or soft-tissue tumor on MRI. Pediatr Radiol 2001; 31:72–75.
8. Youssef BA, Haddad MC, Zahrani A, et al. Osteoid osteoma and osteoblastoma: MRI appearances and the significance of ring enhancement. Eur Radiol 1996; 6:291–296.
9. Helms CA, Hattner RS, Vogler JB III. Osteoid osteoma: radionuclide diagnosis. Radiology 1984; 151:779–784.
10. Spouge AR, Thain LM. Osteoid osteoma: MR imaging revisited. Clin Imaging 2000; 24:19–27.
11. Ehara S, Rosenthal DI, Aoki J, et al. Peritumoral edema in osteoid osteoma on

magnetic resonance imaging. Skeletal Radiol 1999; 28:265–270.
12. Grey AC, Davies AM, Mangham DC, Grimer RJ, Ritchie DA. The 'penumbra sign' on T1-weighted MR imaging in subacute osteomyelitis: frequency, cause and significance. Clin Radiol 1998; 53:587–592.
13. Nogues P, Marti-Bonmati L, Aparisi F, Saborido MC, Garci J, Dosda R. MR imaging assessment of juxta cortical edema in osteoid osteoma in 28 patients. Eur Radiol 1998; 8:236–238.
14. Biebuyck JC, Katz LD, McCauley T. Soft tissue edema in osteoid osteoma. Skeletal Radiol 1993; 22:37–41.
15. Marshall JH, Sonsire JM, Nielsen PE, Nigogosyan G, Terzian J. Digital angiography and osteoblastoma of the triquetrum. J Hand Surg [Am] 1987; 12:256–258.
16. Crouzet G, Mnif J, Vasdev A, Pascal-Ortiz D, Chirossel JP, Pasquier B. Osteoid osteoma of the spine: radiological aspects and value of arteriography—Four cases. J Neuroradiol 1989; 16:145–159.
17. O'Hara JP III, Tegtmeyer C, Sweet DE, McCue FC. Angiography in the diagnosis of osteoid-osteoma of the hand. J Bone Joint Surg Am 1975; 57:163–166.
18. Klein MH, Shankman S. Osteoid osteoma: radiologic and pathologic correlation. Skeletal Radiol 1992; 21:23–31.
19. Levine E, Neff JR. Dynamic computed tomography scanning of benign bone lesions: preliminary results. Skeletal Radiol 1983; 9:238–245.
20. McGrath BE, Bush CH, Nelson TE, Scarborough MT. Evaluation of suspected osteoid osteoma. Clin Orthop 1996; Jun: 247–252.
21. van der Woude HJ, Verstraete KL, Hogendoorn PC, Taminiau AH, Hermans J, Bloem JL. Musculoskeletal tumors: does fast dynamic contrast-enhanced subtraction MR imaging contribute to the characterization? Radiology 1998; 208:821–828.
22. Verstraete KL, Van der Woude HJ, Hogendoorn PC, De-Deene Y, Kunnen M, Bloem JL. Dynamic contrast-enhanced MR imaging of musculoskeletal tumors: basic principles and clinical applications. J Magn Reson Imaging 1996; 6:311–321.